

Divergent and Overlapping Phenotypes in NMOSD and MOGAD: A Literature-Derived Descriptive Synthesis of Reported Clinical, Radiological and CSF Features

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01 ABSTRACT

Neuromyelitis Optica Spectrum Disorder (NMOSD) and Myelin Oligodendrocyte Glycoprotein Antibody-associated Disease (MOGAD) present with substantially overlapping clinical and radiological features, complicating differential diagnosis despite requiring distinct long-term management. This investigation synthesised 407 quantitative prevalence estimates drawn from more than twenty-five published studies, covering 28 standardised clinical, MRI and cerebrospinal-fluid features reported across 187 NMOSD and 220 MOGAD populations. For every feature, median reported prevalence, interquartile range and contributing study count were calculated and compared descriptively. Populations reported a higher prevalence of longitudinally extensive transverse myelitis, area postrema syndrome, bright spotty spinal lesions, and posterior optic-pathway (chiasm and tract) involvement in NMOSD, and a higher prevalence of ADEM, cortical lesions, optic disc swelling, bilateral optic neuritis and cerebrospinal fluid pleocytosis in MOGAD. Broad clinical categories, such as overall optic neuritis, acute myelitis and brainstem syndrome, showed close overlap between the two diseases. Yet, their more specific anatomical subtypes diverged considerably, suggesting that specific phenotypic configuration may be especially informative. Evidence volume varied widely across features, and several apparent divergences rested on a single contributing study. The findings are presented as a citation-traceable, hypothesis-generating literature map, and may contribute towards future steps involving improvements in diagnostic or classification tools.

02 RESEARCH GAP & OBJECTIVES

RESEARCH GAP

- Published population-level prevalence data are rarely synthesised into a standardised, multidimensional phenotype map that enables systematic comparison across NMOSD and MOGAD.
- Published comparisons of NMOSD and MOGAD are typically feature-by-feature and cohort-specific, rather than assembled into a single, citation-traceable framework.
- Broad syndrome categories such as optic neuritis and myelitis are widely compared, but their specific anatomical and radiological subtypes are rarely synthesised alongside broader manifestations.

RESEARCH OBJECTIVES

- Construct a standardised, citation-traceable database of quantitative NMOSD and MOGAD phenotype estimates.
- Calculate the median reported prevalence, interquartile range and contributing study count for each comparable feature.
- Identify clinical, MRI and CSF features showing the clearest reported divergence and overlap between NMOSD and MOGAD populations.
- Integrate evidence volume and between-study heterogeneity into feature-level interpretation to distinguish more consistently reported patterns from findings that may be sensitive to study design, population composition or reporting practices.

- Between-study heterogeneity in clinical context, denominator type and population composition is rarely reported alongside prevalence comparisons, making it difficult to distinguish consistent phenotype patterns from findings influenced by study design or reporting differences.
- Detect and exclude features affected by unit inconsistencies, harmonisation errors, insufficient reporting or incompatible definitions before they enter the primary analysis.
- Use the resulting population-feature profiles to characterise multidimensional phenotypic similarity and differences between NMOSD and MOGAD.

03 METHOD

Evidence Collection

One row was extracted per independently reported quantitative estimate, since each publication frequently reports several distinct prevalence values across populations, timepoints or subgroups. Values were never averaged, pooled or inferred; only percentages explicitly stated in the source text were recorded. Population, clinical context and denominator type were preserved alongside each value. This approach preserves the traceability needed to interrogate downstream inconsistencies.

Standardisation & Harmonisation

Using RStudio, clinical and radiological terminology found in the original literature text were mapped to a fixed set of standardised features, but only where source definitions were genuinely equivalent. Closely related constructs, for example, optic neuritis versus bilateral optic neuritis, were kept distinct to retain clinically meaningful anatomical and severity information. Each estimate was additionally tagged with disease state, study identifier and denominator type, which allowed the primary analysis to be restricted to patient-level, prevalence-type estimates only, excluding diagnostic-accuracy measures such as sensitivity and specificity that numerically resemble, but conceptually differ from, prevalence.

Descriptive Comparative Analysis

Descriptive data from all studies: median, interquartile range and study count, were calculated per feature and population, and percentage-point differences were reported in place of manufactured effect sizes such as risk ratios. Formal statistical significance testing was withheld; treating heterogeneous aggregate percentages as independent patient-level observations would misrepresent the underlying evidence structure. This mirrors established practice in narrative and semi-quantitative evidence synthesis, where transparency about evidence volume is prioritised over spurious statistical precision.

Biological Interpretation & Quality Control

Every comparison was biologically analysed using existing literature identified through an independent bibliographic review. Row counts were reconciled across all derived summary tables to confirm internal consistency, and relationships that were structurally implausible. During quality control, we identified and excluded one feature, serum MOG-IgG seropositivity, after determining invalid quantitative constructs.

Figure 5. Analytical pipeline summary.



04 RESULTS

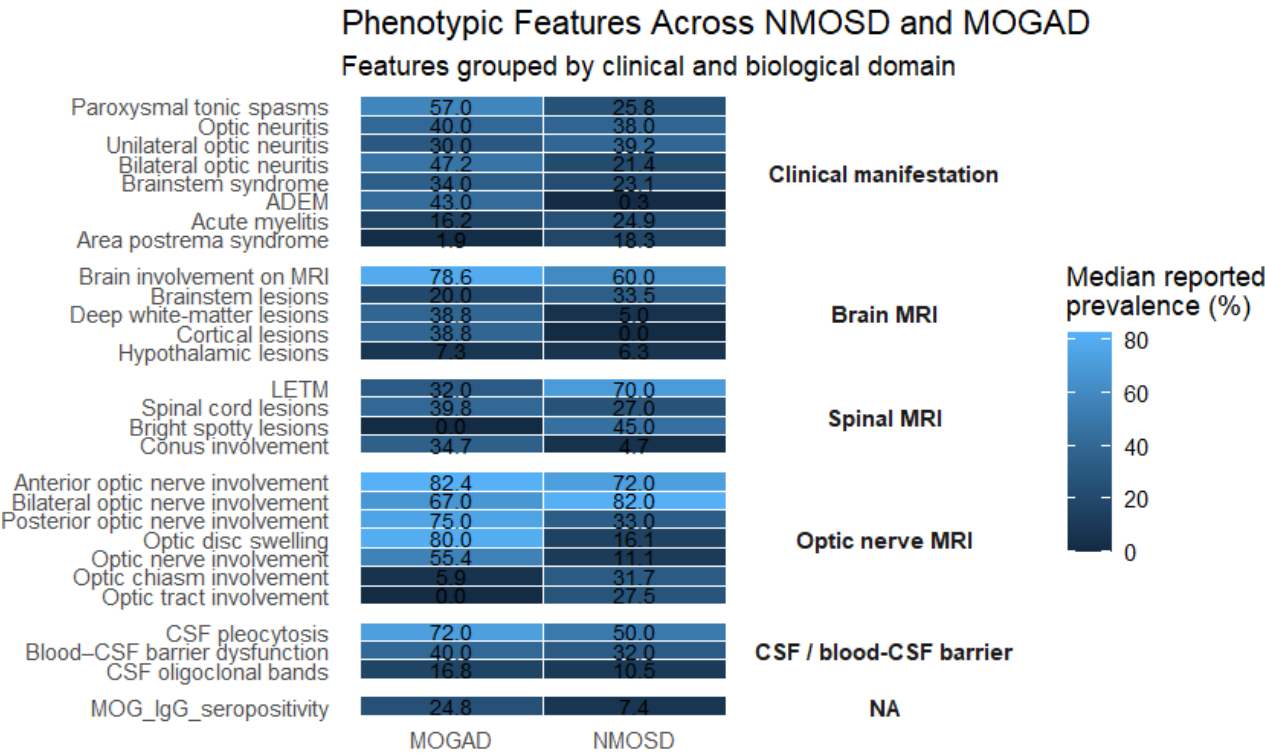


Figure 1. Median reported prevalence (%) of clinical, MRI and CSF features across NMOSD and MOGAD, grouped by domain. Serum MOG-IgG seropositivity is shown for completeness but was excluded from the primary comparison (see Limitations).

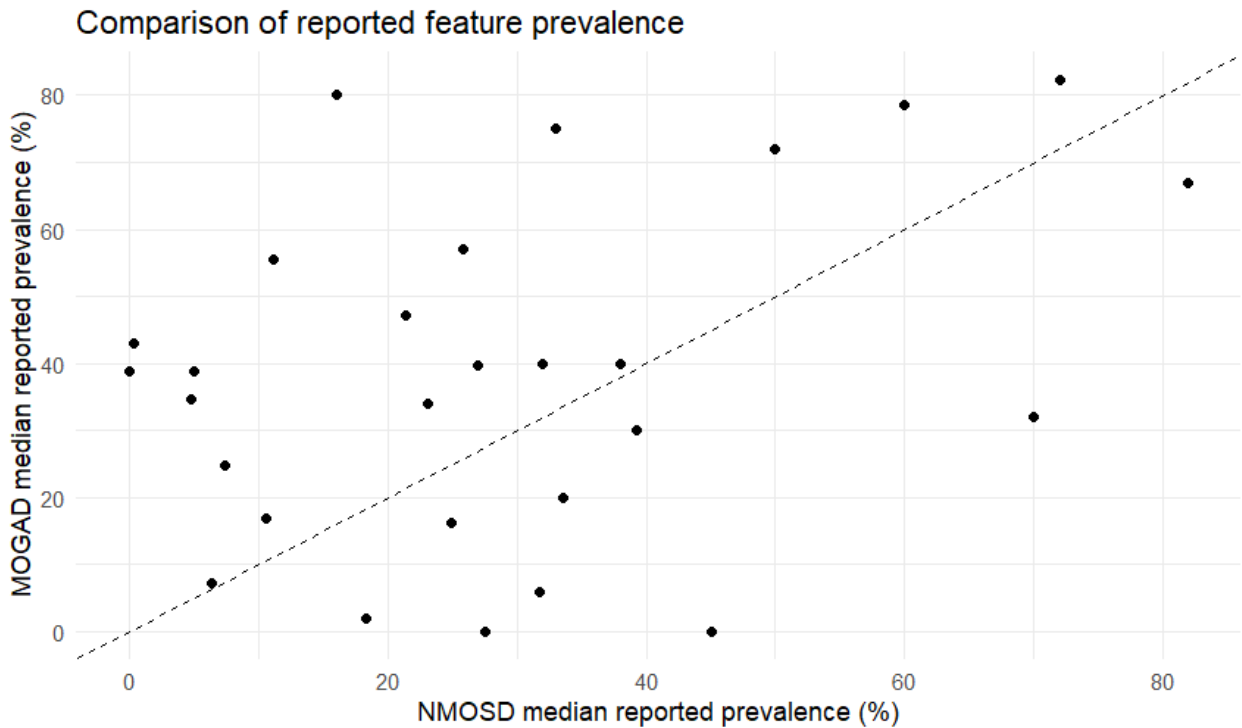


Figure 2. Median reported prevalence in NMOSD (x-axis) plotted against MOGAD (y-axis) for the 28 primary comparable features; the dashed line marks equal reported prevalence.

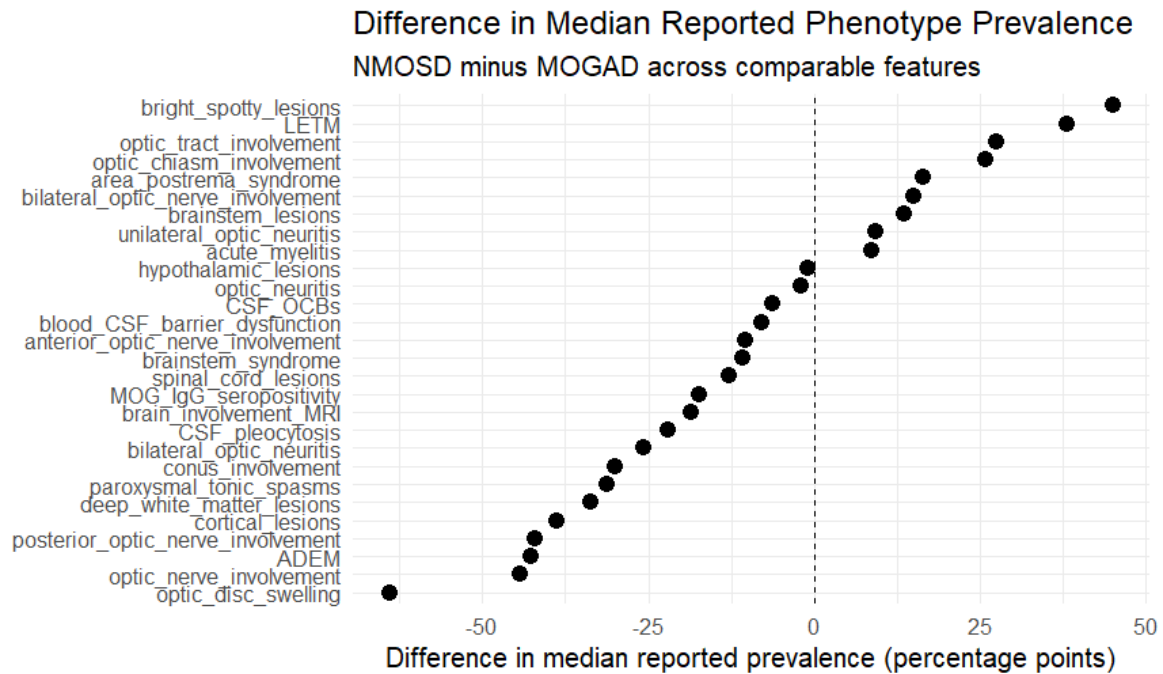


Figure 3. Ranked percentage-point differences in median reported prevalence (NMOSD minus MOGAD) across the 28 comparable features.

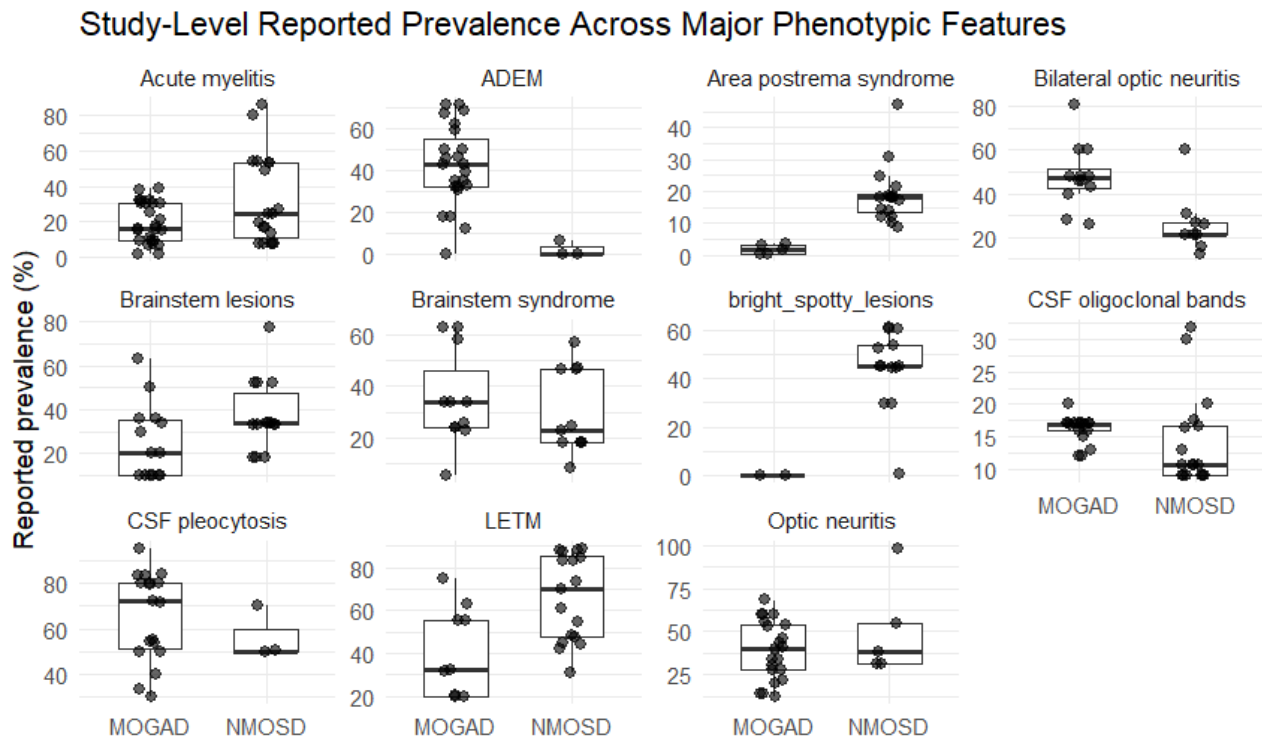


Figure 4. Study-level distribution of reported prevalence for the ten most evidence-dense shared features, showing the between-study variation underlying each summary median.

Twenty-eight standardised features were eligible for direct NMOSD–MOGAD comparison after excluding serum MOG-IgG seropositivity. Table 1 summarises the sixteen features with the most decisive reported divergence or the broadest evidence base; the complete 28-feature comparison, including sparser features, underlies Figures 1–3.

Table 1. Selected NMOSD–MOGAD phenotype comparison (median reported prevalence, %).

Feature	NMOSD (studies)	MOGAD (studies)	Diff. (pp)	Direction
ADEM	0.3 (2)	43.0 (16)	42.7	MOGAD
Bright spotty lesions	45.0 (12)	0.0 (2)	45.0	NMOSD
Optic disc swelling	16.1 (2)	80.0 (3)	64.0	MOGAD
LETM	70.0 (14)	32.0 (5)	38.0	NMOSD
Cortical lesions	0.0 (3)	38.8 (6)	38.8	MOGAD
Deep white-matter lesions	5.0 (7)	38.8 (4)	33.8	MOGAD
Optic tract involvement	27.5 (2)	0.0 (1)	27.5	NMOSD
Bilateral optic neuritis	21.4 (8)	47.2 (9)	25.8	MOGAD
Optic chiasm involvement	31.7 (5)	5.9 (3)	25.8	NMOSD
CSF pleocytosis	50.0 (2)	72.0 (12)	22.0	MOGAD
Area postrema syndrome	18.3 (11)	1.9 (3)	16.4	NMOSD
Brainstem lesions	33.6 (10)	20.0 (6)	13.6	NMOSD
Spinal cord lesions	27.0 (3)	39.8 (2)	12.8	MOGAD
Brainstem syndrome	23.1 (8)	34.0 (10)	10.9	MOGAD
Acute myelitis	24.9 (13)	16.2 (14)	8.7	NMOSD
Optic neuritis (overall)	38.0 (5)	40.0 (16)	2.0	Overlap

Across the full comparable set, NMOSD populations reported the highest relative prevalence for longitudinally extensive transverse myelitis, bright spotty spinal lesions, area postrema syndrome, and posterior optic-pathway involvement (chiasm and tract), several of which are supported by 10 or more contributing studies per disease. MOGAD populations reported the highest relative prevalence of ADEM, cortical and deep white-matter lesions, optic disc swelling, bilateral optic neuritis and CSF pleocytosis, spanning encephalitic, optic and CSF domains rather than a single anatomical system. Broad syndrome-level categories, including optic neuritis overall, acute myelitis, brainstem syndrome, hypothalamic lesions and CSF oligoclonal bands, showed comparatively little separation between the two diseases, with differences generally under 11 percentage points. Figure 4 illustrates that even the most evidence-dense features carry considerable between-study spread.

05 FINDINGS

Finding 01 — A Spinal–Brainstem–Posterior Optic Signature Characterises NMOSD

NMOSD populations reported a coherent cluster of features anchored in spinal cord and posterior visual-pathway involvement: longitudinally extensive transverse myelitis (70% vs 32%), bright spotty spinal lesions (45% vs 0%), area postrema syndrome (18.3% vs 1.9%), and involvement of the optic chiasm (31.7% vs 5.9%) and optic tract (27.5% vs 0%). Several of these directions have ten or more contributing studies per disease, giving this cluster comparatively strong evidentiary weight relative to other divergent features.

Mechanistically, this pattern maps onto NMOSD's identity as an astrocytopathy. AQP4-IgG targets water channels densely expressed on astrocytic end-feet, which cluster most heavily in the spinal cord, optic nerve, and circumventricular organs. The area postrema is particularly vulnerable because it lacks a tight blood–brain barrier, allowing circulating antibody direct access — consistent with both the high reported frequency of area postrema syndrome and its typically favourable recovery. The longitudinally extensive lesion pattern reflects the contiguous distribution of AQP4 along the spinal cord, while bright spotty lesions are thought to

represent necrotic and microcystic change from severe astrocytic injury. Posterior optic-pathway involvement (chiasm and tract) is similarly explained by astrocyte localisation within these structures, distinguishing NMOSD sharply from MOGAD's anterior-predominant pattern.

Finding 02 — MOGAD's Signal Is a Broader, Multi-System Phenotype

MOGAD's divergent features span several distinct biological compartments rather than one anatomical system. ADEM (43% vs 0.3%), cortical lesions (38.8% vs 0%) and deep white-matter lesions (38.8% vs 5%) together suggest an encephalitic pattern. Optic disc swelling (80% vs 16.1%) and bilateral optic neuritis (47.2% vs 21.4%) describe a distinctive optic-neuritis phenotype. Conus involvement (34.7% vs 4.7%) contributes a separate spinal-distribution signal. Independent cohorts support these directions (Gowda et al., 2025; Satukijchai et al., 2022; Chen et al., 2024), though the ADEM comparison in particular rests on only two NMOSD studies set against sixteen MOGAD studies, and thus, may partly reflect the disproportionate representation of paediatric MOGAD cohorts in the literature.

This multi-compartment signal is consistent with MOGAD's underlying pathology as an oligodendropathy. MOG-IgG and CD4⁺ T cells drive perivenous inflammation concentrated around small veins in deep and subcortical white matter and deep cortical layers, producing the cortical and encephalitic features seen here and in ADEM. The anterior, bilateral optic-nerve pattern reflects the distribution of MOG antigen along the orbital nerve segment and its access route via the blood-retinal barrier, in contrast to NMOSD's posterior predilection. The elevated conus involvement is a further anatomical signature: the conus contains comparatively little white matter but a high surface-to-volume ratio, which may increase its relative vulnerability in MOG-IgG-mediated disease.

Finding 03 — Broad Syndrome Labels Understate the True Divergence

Reported prevalence of optic neuritis overall (38% NMOSD vs 40% MOGAD — a two-percentage-point difference) and of acute myelitis (24.9% vs 16.2%) showed close overlap between the two diseases. Yet the anatomical subtypes underlying each syndrome diverged sharply: bilateral involvement, disc swelling, and chiasmal or tract involvement each differed by 22 to 64 percentage points, and a similar pattern separates overlapping brainstem syndrome from more divergent brainstem lesion counts. This suggests that broad syndromic labels can obscure distinctions that only become visible once the underlying anatomical configuration is examined feature by feature — a pattern that motivates moving from single-feature comparison toward a multidimensional phenotype representation in future work.

Biologically, this pattern is expected: both diseases can affect the optic nerve and spinal cord, converging on the same clinical syndrome labels, but via different antigenic targets with different lesion topography. In the optic nerve, NMOSD's astrocytopathy favours the posterior segment and chiasm, while MOGAD's oligodendropathy favours anterior, longitudinally extensive, bilateral involvement. In the spinal cord, NMOSD produces centrally located LETM with necrotic bright spotty change, while MOGAD lesions can extend broadly but show a particular predilection for the conus. Broad syndrome prevalence therefore conceals a fundamental anatomical difference that only becomes visible once features are disaggregated.

Finding 04 — A Minority of Apparent Differences Rest on Fragile, Single-Study Evidence

Several apparent divergences were supported by only one contributing study per disease, including posterior optic nerve involvement (33% vs 75%) and paroxysmal tonic spasms (25.8% vs 57%). A structural inconsistency was also identified: reported NMOSD longitudinally extensive transverse myelitis (70%)

numerically exceeded reported NMOSD spinal cord lesions overall (27%), a relationship that cannot hold within one consistent patient population and instead indicates that the two medians were derived from non-overlapping sets of studies with different denominators and contexts

The paroxysmal tonic spasms figure is biologically implausible as a general MOGAD feature: PTS is mechanistically linked to ephaptic transmission during recovery from severe, longitudinally extensive spinal cord damage, which is a hallmark of NMOSD rather than MOGAD. The LETM/spinal-cord-lesion inconsistency is similarly not a biological signal: as LETM is by definition a subset of spinal cord lesions, a median LETM prevalence exceeding overall spinal cord lesion prevalence is a methodological artefact of pooling non-equivalent literature samples.

06 LIMITATIONS

Evidence volume is highly unequal across features: longitudinally extensive transverse myelitis is supported by 14 NMOSD studies against 5 for MOGAD, while ADEM is supported by only 2 NMOSD studies against 16 for MOGAD. The apparent magnitude of a reported difference cannot therefore be read independently of how much evidence underlies each side, and several of the largest percentage-point differences in this dataset are also among the most sparsely supported.

The raw extraction contained repeated study–population–feature rows, such as duplicated ADEM, CSF oligoclonal band and LETM entries attributable to a small number of sources, which may have inflated apparent evidence counts ahead of full deduplication.

Reported estimates combine several distinct clinical contexts (onset, first attack, acute attack, MRI examination, follow-up and overall disease course) that are not necessarily interchangeable and may differ in terms of prevalence. Omitting consideration of such clinical contexts may obscure meaningful differences in reported feature prevalence and introduce apparent heterogeneity that reflects differences in measurement timing rather than underlying phenotypic variation.

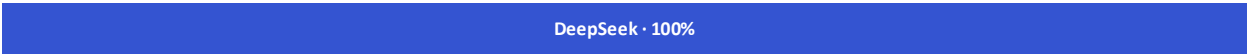
Cohort composition (age, geography, AQP4/MOG serostatus and diagnostic era) was not uniformly reported across sources. No dedicated AQP4-negative NMOSD subgroup could be isolated, and paediatric MOGAD status was inconsistently flagged; both factors may confound disease-level comparisons, particularly the large reported ADEM difference.

Several features rest on a single contributing study for one or both diseases and should be read as hypothesis-generating rather than stable disease-level estimates. No formal meta-analysis, pooled effect size, confidence interval or significance test was calculated at any stage; all comparisons reported here are descriptive percentage-point differences across heterogeneous literature, and imaging-derived features in particular remain susceptible to ascertainment differences between studies designed around MRI protocols and those drawing on general clinical cohorts.

07 AI INTEGRATION

The table below documents which AI systems contributed to each stage of this investigation and their approximate relative share of use within that stage. Bars are proportional to reported share; stages are not directly comparable to one another in absolute effort.

Data Extraction



Data Processing



Computational Analysis Instructions (execution in RStudio)



Results Interpretation



Literature Validation



■ DeepSeek ■ ChatGPT ■ Claude

08 CONCLUSION

This synthesis assembled 407 literature-derived prevalence estimates into a standardised, citation-traceable comparison of 28 clinical, radiological and cerebrospinal-fluid features across NMOSD and MOGAD. The clearest reported divergences describe two distinguishable phenotypic clusters: a spinal–brainstem–posterior optic pattern in NMOSD, comprising longitudinally extensive transverse myelitis, bright spotty spinal lesions and area postrema syndrome; and a broader encephalitic and optic-nerve pattern in MOGAD, comprising ADEM, cortical and deep white-matter lesions, optic disc swelling and bilateral optic neuritis. Broad syndrome-level categories, by contrast, showed close overlap, suggesting that specific anatomical configuration carries more discriminating information in the published literature than syndrome label alone.

These findings should be read as a descriptive map of the existing published evidence: no meta-analytic pooling, statistical significance testing, or individual-level classification was performed, and several flaws in the data, e.g. serum MOG-IgG seropositivity, paroxysmal tonic spasms, and the internally inconsistent NMOSD LETM/spinal-cord-lesion relationship, required explicit flagging and could not be biologically interpreted. Evidence volume varied substantially across features, and the dataset's shortfall against originally anticipated estimate counts remains only partially reconciled.

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